ITM University Gwalior

School of Computing and Intelligent Systems

Project-Based Learning (PBL)

Synopsis Approval Submission Format

Course: Data Analytics using Python
Course Code:CSL-0312
Program:B.Tech. CSE(AIML)
Semester:III
Academic Session: July-Dec 2026

PROJECT-BASED LEARNING (PBL) TOPIC APPROVAL FORM

1. Student Details

Team Leader Name& Roll No.: Rishu Gupta (BETN1AI25110)

Team Member Name& Roll No.: Vinit Yadav (BETN1AI25084)

Team Member Name& Roll No.: Sudhanshu Kumar (BETN1AI25109)

Branch/Program:B.Tech AI & ML

Semester & Section:3rd SEM , "E"

Faculty Mentor: Dr Sanjay Jain

Date of Submission: 11/8/2026

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2. PBL Topic Title

Domain Name : Healthcare

Proposed Title:Disease Frequency Analysis Using Indian Healthcare Records with Python

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3. Problem Statement / Brief Overview

Description:

Across India, patients are being admitted to hospitals with various diseases, and hospitals are collecting large amounts of data. However, despite having so much data, we are not using it effectively.

For example, we should be able to analyze the data to understand which diseases are becoming more common, which age groups are most affected, and which conditions are more prevalent among children, women, and senior citizens. We should also identify diseases associated with different BMI ranges and evaluate which treatments produce the best outcomes.

If we analyze this data correctly and share the insights with the public, people can become more aware, avoid common mistakes, and take preventive steps to protect themselves when they fall into a higher-risk category.

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4. Objectives of the Project

1) We will analyze healthcare data to generate insights, such as how frequently each disease occurs and which diseases are the most common.

2) To identify the relationship between diseases and demographic factors such as age , gender and region.

3) We will understand the patterns of each common disease—for example, which age groups are at higher risk and which gender should take greater preventive measures against a particular disease.

4) To identify trends and variations in disease occurrence across different regions and demographic groups using the available healthcare data.

5) To present the key finding in an interactive and easy-to-understand dashboard , allowing users to explore disease patterns using relevant filters such as disease , age , gender and region.

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5. Dataset Details

Dataset Name: Indian Healthcare Patient Records

Source/Organization: Keggle Datasets

Dataset Link (URL):https://www.kaggle.com/datasets/arunsworkspace/indian-healthcare-patient-records

Type of Dataset:☑ Public ☐ Private ☐ Self-Collected

Dataset Format: CSV

Approximate Number of Records:around 550 – 650 rows

Number of Features/Attributes: 19

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6. Proposed Methodology / Work Plan

Phase Proposed Work

Phase 1 :- Dataset Collection and Understanding

Phase 2 :- Data Cleaning and Preprocessing

Phase 3 :-Exploratory Data Analysis (EDA)

Phase 4 :- Data Visualization using Python Libraries and dashboard development using Power BI

Phase 5 :- Insight Generation and Interpretation

Phase 6 :- Report Preparation, Presentation, and GitHub Documentation

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7. Software, Tools and Technologies

Mention the software, programming languages, libraries, frameworks, and development tools to be used.

- Programming Language(s): Python
- Python Libraries: Pandas, NumPy, Matplotlib, Seaborn
- IDE/Development Environment: Google Colab
- Other Tools/Software: Power BI and Streamlit

8. Expected Deliverables

- ☑ Python Source Code
- ☑ Google Colab Notebook
- ☑ Cleaned Dataset
- ☑ Data Visualizations
- ☑ Dashboard
- ☑ Project Report
- ☑ Presentation (PPT)
- ☑ Demonstration/Prototype

9. Expected Outcomes

The project is expected to provide meaningful insights into disease frequency, patient demographics, treatment outcomes, and healthcare trends using real-world healthcare data. It will demonstrate practical skills in data cleaning, preprocessing, exploratory data analysis, and data visualization. The analysis will help identify patterns that can support better healthcare planning and decision-making.

10. Innovation

This project goes beyond traditional data analysis by developing an interactive healthcare dashboard using Power BI. Instead of presenting only static charts and graphs, the dashboard allows users to filter data based on region, disease, gender, hospital type, and other healthcare parameters. The visualizations are designed to be simple and easy to understand, enabling even non-technical users, such as hospital administrators and healthcare staff, to interpret the data without requiring programming or data analysis knowledge. The project transforms raw healthcare data into meaningful insights, making data-driven decision-making faster, more effective, and user-friendly.

11. References

1. Original Indian Healthcare Dataset (Kaggle / Original Source)
2. Pandas Documentationhttps://pandas.pydata.org/docs/
3. NumPy Documentation https://numpy.org/doc/
4. Matplotlib Documentationhttps://matplotlib.org/stable/

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12. Declaration by Student

I hereby declare that the proposed Project-Based Learning (PBL) topic is my/our original work. The dataset and references mentioned above are authentic to the best of my/our knowledge. I/We understand that any change in the approved topic will require prior approval from the faculty mentor.

Students Signature: Disha Gupta , Ornat. , Sudhanshu Kr

Date: 11/08/2026

Submission Guidelines

1. The proposed topic must be relevant to the course outcomes.
2. Use only authentic and reliable datasets (e.g., UCI Machine Learning Repository, Kaggle, Data.gov, Government Open Data Portals, etc.).
3. Clearly define the problem statement and project objectives.
4. Mention the complete dataset source link.
5. The proposed work should be feasible to complete within the semester duration.
6. Cite all references properly to avoid plagiarism.
7. Submit this approval form before starting project implementation.
8. Any modification to the approved topic or dataset requires prior approval from the faculty mentor.

PBL Synopsis Faculty Evaluation

Team Leader (Roll No.):	Rishu Gupta (BETN1AI25110)
Team Members:	Vinit Yadav (BETN1AI25084), Sudhanshu Kumar (BETN1AI25109)
Proposed Title:	Disease Frequency Analysis Using Indian Healthcare Records with Python
Course:	Data Analytics using Python (CSL-0312), B.Tech CSE(AIML), Semester III

Evaluation Criteria	Marks	Remarks (as per this submission)
Relevance of Topic (out of 1.5)	1.4	Healthcare analytics topic is relevant and practically motivated.
Problem Statement (out of 1.5)	1.3	Problem statement clearly explains the gap between large hospital data volumes and their under-utilization for public awareness.
Availability of Dataset (out of 1.5)	1.3	Kaggle 'Indian Healthcare Patient Records' dataset, ~550-650 rows, 19 features - modest in size but adequate for the stated EDA-level analysis, and correctly linked.
Feasibility (out of 1.5)	1.3	6-phase plan (collection -> cleaning -> EDA -> Power BI/dashboard visualization -> insight generation -> reporting) is complete and realistic.
Scope of Work (out of 1.5)	1.3	Scope (disease frequency by demographic group, using Power BI dashboarding in addition to Python) is a reasonable, achievable extension.
Expected Learning Outcomes (out of 1.5)	1.2	5 objectives are specific and each maps to a phase.
Innovation (out of 1)	0.9	The use of an interactive Power BI dashboard (in addition to standard Python EDA) specifically to make findings accessible to non-technical hospital staff is a genuine, well-explained innovation.
Total	**8.7 / 10**	

Overall Recommendation: ☒ **Approved** ☐ Approved with Minor Modifications ☐ Revision Required ☐ Not Approved

Faculty Remarks:

Complete and well-justified submission. Approved as submitted.

Faculty Mentor Signature: _____

Date: 1/9/26